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CHAPTER 1: INTRODUCTION

1.1 Introduction

A Job portal also known as a career portal, is a modern name for an online job board that helps applicants find the jobs and aids employers in their quest to locate ideal candidates as per their experiences, qualification and so on. The system is to provide an alternate and convenient way for a person to find easily. Some government agencies, nonprofit organizations, universities, private business have their own job portals that applicants can access on the organization’s website. [1]. A job portal is a website where people who need jobs can find jobs and companies looking for job seekers can find the perfect employees. It serves as a medium where job seekers can create profiles, upload resumes, and search for job openings across various industries and locations. Employers, on the other hand, can post job vacancies, review applicant profiles, and conduct the initial stages of the hiring process through the portal. A well-known job portal is LinkedIn. It lets professionals show what they’re good at, connect with others in their field, and check out job listings from companies. It’s a big platform for making work connections and finding jobs.[2]

1.2 Problem Statement

In today’s dynamic job market, job seekers often face difficulties in identifying roles that truly match their skills, qualifications, and preferences. Most conventional job portals rely on manual keyword-based search mechanisms, which can be inefficient and time-consuming. These systems typically lack advanced features to automatically analyze user resumes or provide personalized job recommendations. This limitation results in job seekers browsing through hundreds of unrelated listings, increasing the chances of missing suitable opportunities. At the same time, employers struggle to identify the most relevant candidates from large applicant pools, especially when there's no intelligent filtering based on resume content. Furthermore, fresh graduates and users with less experience often receive generic suggestions, making it harder for them to find entry-level roles that suit their profiles. Therefore, there is a strong need for an intelligent system that can parse resumes, extract meaningful data, and offer job suggestions that are tailored to the user—enhancing the effectiveness and user experience of job portals.

According to a study on recruitment platform limitations, over 60% of users prefer personalized job suggestions over standard search results, indicating a demand for smarter matching systems that consider user data more intelligently [3].

1.3 Objectives

- To develop a user-friendly job portal tailored for Nepali job seekers. considering all the aspects of security.
- To provide automated job recommendations based on parsed resume content using keyword matching and relevance scoring.
- To reduce the time and effort required in manual job searching.

1.4 Scope and Limitations

1.4.1 Scope

Job portal will be developed for solves the problem of manual job searching. We built a resume parser that reads PDF and DOCX files using libraries like pdf.js-extract and mammoth. It extracts important details like name, email, skills, and experience using regular expressions.

We then apply the **Jaccard Similarity** algorithm to compare the extracted text with job descriptions stored in our database. The algorithm returns a match score for each job, which we use to recommend the top relevant jobs to the user.

This system automates the job recommendation process and can be very useful for job portals, especially for freshers or users unsure where to apply

1.4.2 Limitation

This system lacks features such as single user can have multiple profile, ATS cheacker, direct message to the recruiter and worker.

The report has been prepared following the guidelines provided by Tribhuvan University. The report is separated into different chapters. Each chapter consists of various sub chapters with its content. The preliminary section of the report consists of Title Page, Acknowledgement, Abstract, Table of Contents, List of Abbreviations. The main report is divided into 5 chapters, which include:

1. Chapter 1: Introduction

It includes the general overview of the system and the project as a whole. It includes the Problem Statement, Objectives, Scope/Limitations and the Development Methodology for the project and the system being developed.

2. Chapter 2: Background Study and Literature Review

It includes the study of the current scenario/environment the system will be deployed into. It includes the study of the current trends, preferences of people, the existing systems, areas of improvement among others.

3. Chapter 3: System Analysis and Design

It includes the requirement and feasibility analysis of the system that can be generated through the studies presented in the previous two chapters. It will also include the Flowchart, ER and DFD for the system which specifies the workflow, entities, attributes and their relationships. It includes the design of the database, forms and interface of the system. It also includes the implementation details of the selected methodology and the details of the algorithm used.

4. Chapter 4: Implementation and Testing

It includes the details of the different design and development tools used and the implementation details of the modules presented in the form of code snippets of functions, classes. It also includes the testing of the system with different test cases as per the requirement.

6. Chapter 6: Conclusion and Future Recommendations

It includes the summary of the system and the project as a whole. It also includes the possibilities/aspects which the system can implement in the future.

CHAPTER 2:

BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background Study

Traditional Approach

Traditionally, job search and recruitment have relied heavily on manual and offline methods. Job seekers would depend on newspaper advertisements, word-of-mouth, job fairs, and personal visits to companies to find employment opportunities. Employers, on the other hand, posted vacancy notices on bulletin boards or newspapers and collected physical resumes. This approach was often time-consuming, limited in reach, and inefficient. The lack of centralized information made it difficult for both employers and candidates to quickly connect, resulting in prolonged hiring cycles and missed opportunities.

Digital Approach – Job Portals

With the advancement of internet technology and widespread digital adoption, the recruitment landscape has shifted significantly toward online platforms. Job portals serve as digital marketplaces where employers can post job vacancies, and job seekers can create profiles, upload resumes, and apply for jobs instantly. These platforms automate and streamline the recruitment process, offering several advantages over traditional methods, including wider reach, faster communication, and intelligent job matching algorithms.

Modern job portals often incorporate features like resume parsing, keyword extraction, and personalized job recommendations, improving the accuracy of matching candidates to suitable jobs. Additionally, real-time notifications and tracking of application status enhance the overall user experience. The digital approach also enables data-driven insights for employers and supports secure and transparent interactions between stakeholders.

This project aims to build a job portal tailored to the specific needs of the Nepali job market, leveraging the benefits of digital recruitment while addressing local challenges such as language preferences and limited digital literacy. By transitioning from the traditional recruitment practices to an efficient, technology-driven platform, the portal intends to facilitate better employment connections and contribute to reducing unemployment in the region.

2.2 Literature Review

The authors in a research paper on the topic “Notes-Taking Apps with Security Features” had mentioned that Smartphone applications (apps) provide users with features to maximize the usefulness of smart- phones in various categories, such as finance, education, health, life, and entertainment. For these features, apps store within themselves user data, which are closely related to their user. Such data can be thus used as key digital forensics clues. However, some apps use their own security features to protect data against external threats. Security features, which can effectively protect sensitive data, impose considerable digital forensics challenges that require data decryption to be used as evidence. Therefore, it is essential to conduct a preliminary study of apps with security features so that foren-sic investigators can perform their work efficiently.

The authors in a research paper on the topic “Review on Existing Notes-Taking and Task-Management Apps” highlighted that the constantly changing universe of note-taking and task-management applications has become an essential aspect of our digital life. This evaluation delves into the complexities of current programs, revealing their features, strengths, and limits. This study intends to aid users in making educated decisions about selecting an app adapted to their specific needs by critically analyzing the present status of these tools. Note-taking and task-management software play a critical role in organizing and optimizing individual and communal efforts in the changing world of digital productivity tools. This in-depth examination aims to delve into the detailed features, user experiences, and technology landscapes of existing note-taking and task-management applications. This evaluation seeks to provide users and developers with a detailed perspective of the emerging digital productivity ecosystem by scrutinizing the strengths and limits of popular platforms.

CHAPTER 3: SYSTEM ANALYSIS AND DESIGN

3.1 System Analysis

Systems analysis the process of observing systems for troubleshooting or development purposes. It is applied to information technology, where computer-based systems require defined analysis according to their makeup and design. [2]

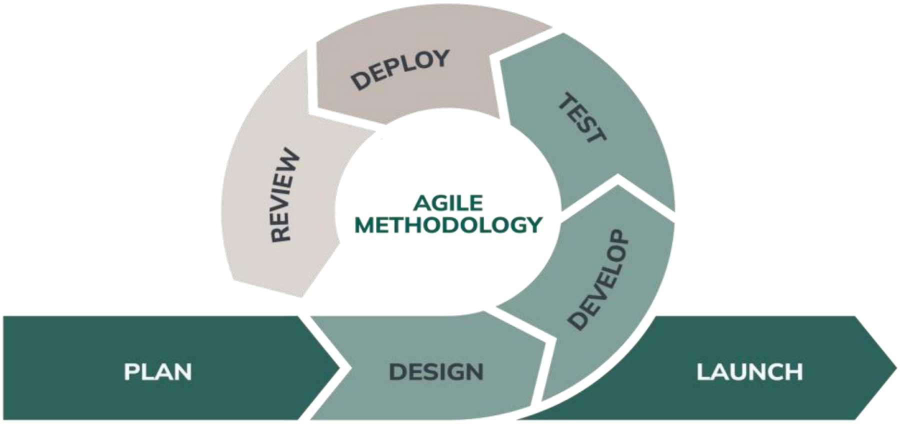


Figure 3.1: Agile Model for Job Portal Web Application

3.1.1. Requirement Identification and Planning

This web application needs to fulfill the given functional and non-functional requirements:

Functional Requirements:

- The application should be able to register and login by users.
- The application should provide access to the user accounts with their valid login credentials.
- The application should be able to see the job and apply the job and get notified.
- The application should be able to add the company and edit company detail.
- The application should be able to upload resume.

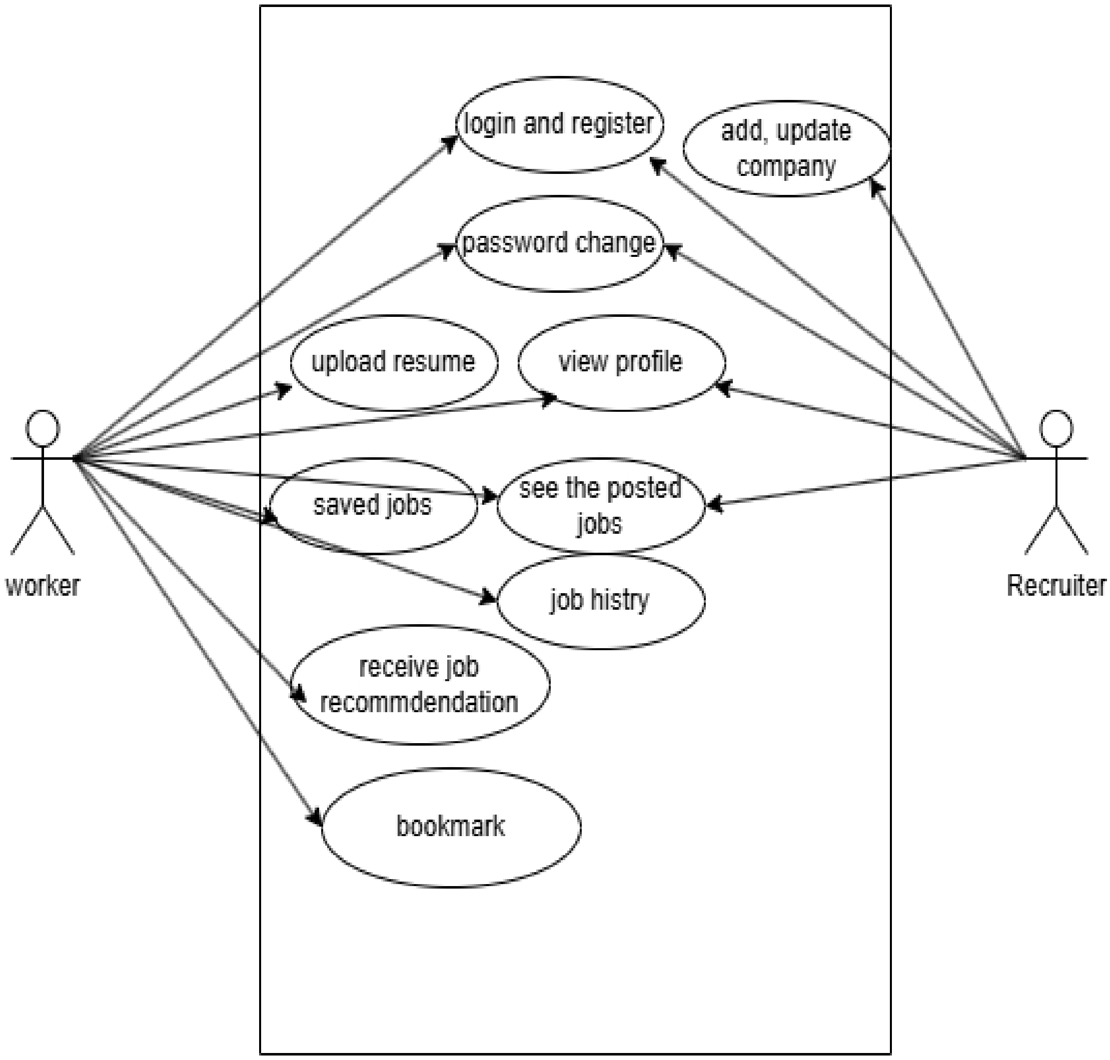


Figure 3.2: Use case diagram for Job portal Web Application

Non-Functional Requirement:

- The application should consider all the aspects of security.
- The application should have user-friendly interface.
- The application should be better in performance.
- The application should be interactive user interface.

3.1.2 Feasibility Analysis

A feasibility study, also known as feasibility analysis, is an analysis of the viability of an idea. It describes a preliminary study undertaken to determine and document a project's viability. The results of this analysis are used in making the decision whether to proceed with the project or not. In short, a feasibility analysis evaluates the project's potential for success, following feasibility analysis was performed prior to working on the project:

I. Technical Feasibility Study

This project can be considered technically feasible as it deals with the secure storage of users notes and it can be easily implemented using the available technologies.

ii. Operational Feasibility Study

This project can be a user-friendly application with no extra knowledge required for its operation. It would be as simple as saving some sort of important data.

Iii. Economic Feasibility Study

There is no much expense in developing this project as the technologies required for it are mostly open-source and easily available to use.

iv. Schedule Feasibility Study

The project expands over a period of 88 days with each tasks being carried out step by step following the system development model. It is completed within scheduled time and do not exceed the scheduled time.

Table 3.1: Gantt chart table for Job Portal

Task Name	**Duration**
Research and Planning	4 days
Requirement Gathering	4 days
System Design	10 days
System Development	40 days
Review	20 days
Testing	10 days

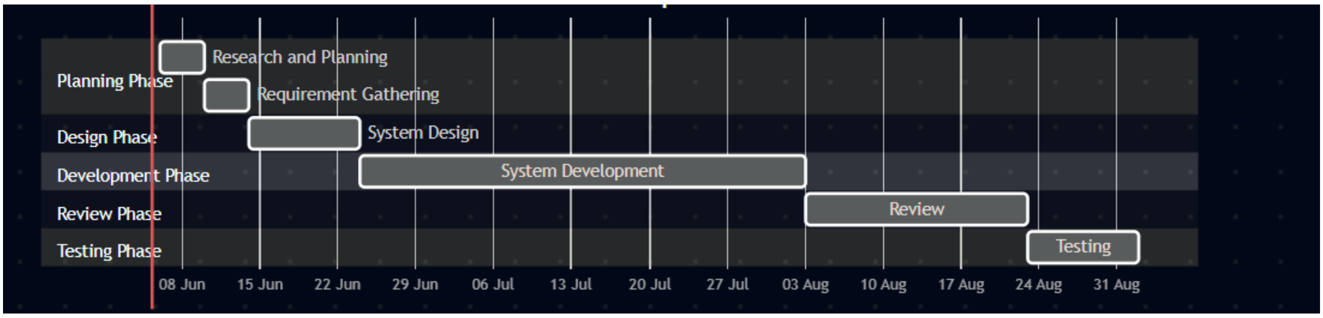
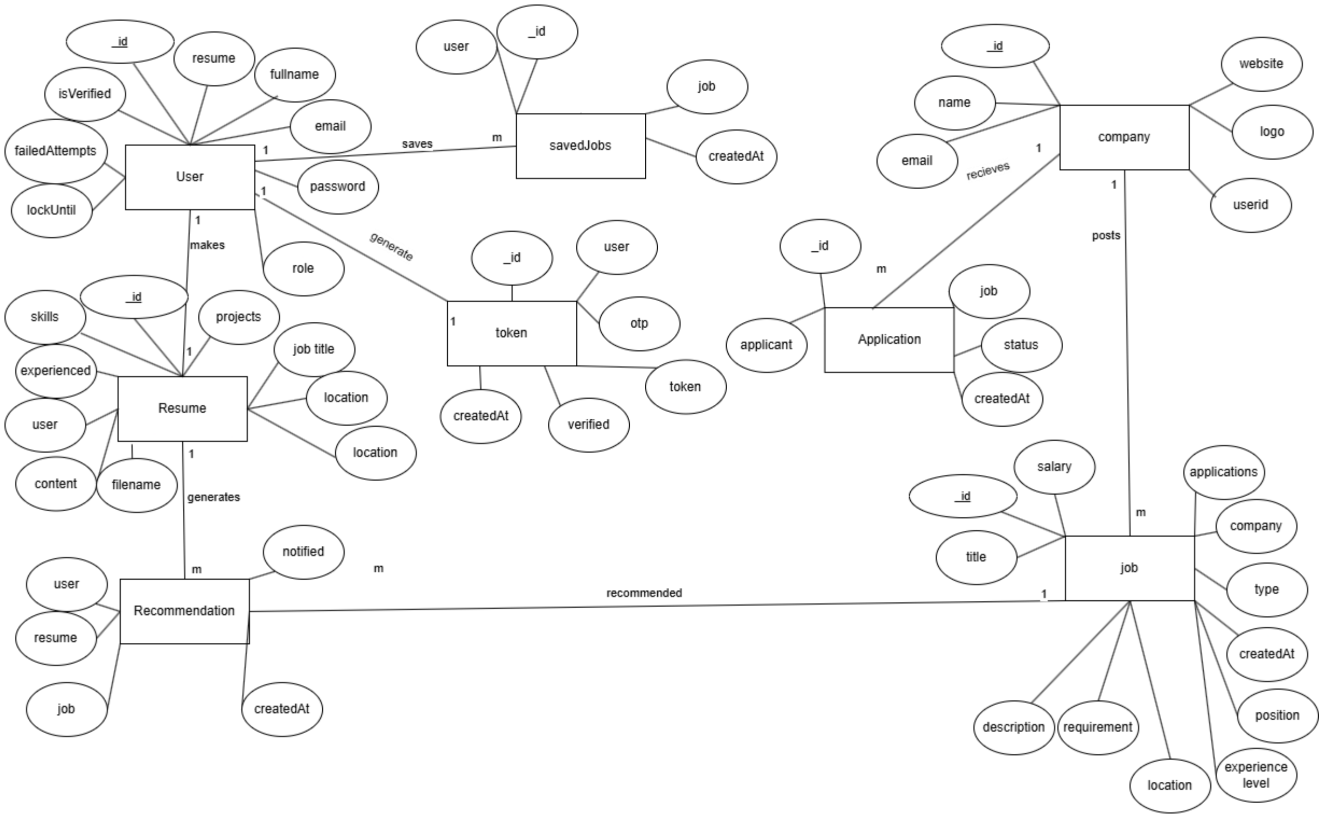


Figure 3.3: GANTT chart for job portal Web Application

3.1.3 <u>Data Modeling (ER-Diagram)</u>



<u>Figure 3.4: ER-Diagram for Job Portal Web Application</u>

The user entity has attributes such as _id, name, email, and password, where _id is the primary key used to uniquely identify each user. Other profile-related fields like profilePic, role, and phoneNumber are also part of this entity. This entity is central to both job seekers and recruiters.

The resume entity has attributes such as _id, content, skills, jobTitles, location, and user, where user is a foreign key that references the user entity. This entity stores extracted information from uploaded resumes and supports the job recommendation logic.

The application entity has attributes like _id, user, job, status, and appliedAt. Here, both user and job are foreign keys referring to their respective entities. This entity is used to track which user has applied to which job and the current status of that application.

The savedjob entity contains attributes such as _id, user, job, and savedAt. Both user and job are foreign keys. This table helps to manage jobs that users have bookmarked for later reference or application.

The recommendation entity includes _id, user, resume, job, score, and matchingReason. Here, user, resume, and job are foreign keys pointing to their respective entities. This entity holds AI-generated job recommendations based on the resume content.

The company entity includes attributes such as _id, name, email, website, logo, and userId, where userId is a foreign key referring to the recruiter who manages the company profile. This allows linking a recruiter account with a company.

The job entity has attributes like _id, title, description, requiredSkills, location, type, experienceLevel, salary, and company. The company field is a foreign key pointing to the company that posted the job.

Lastly, the token entity includes _id, token, otp, and user, where user is a foreign key referencing the user entity. This entity is used for managing email verification and password reset workflows securely

3.1.4 Process Modeling (DFD)

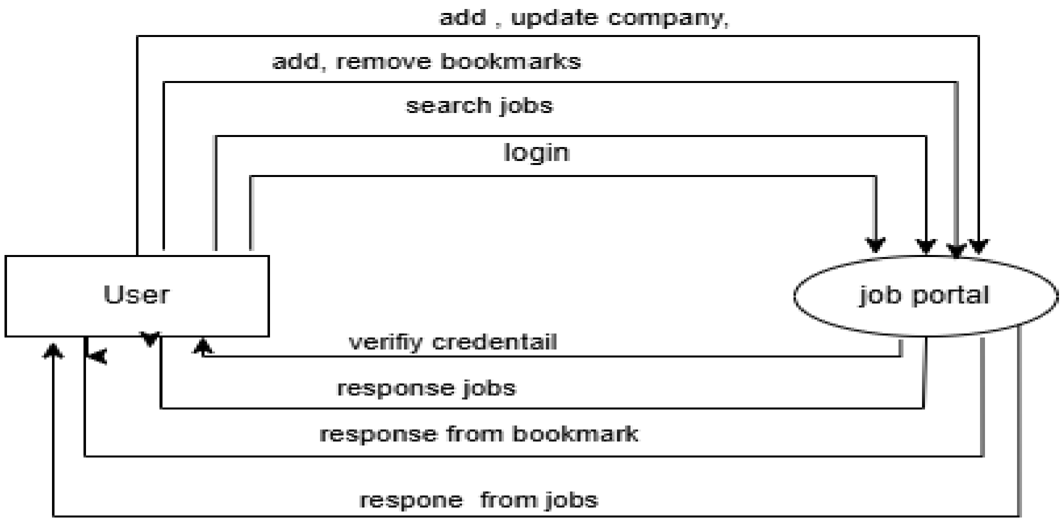


Figure 3.5: Level 0 DFD for Job Portal Web Application

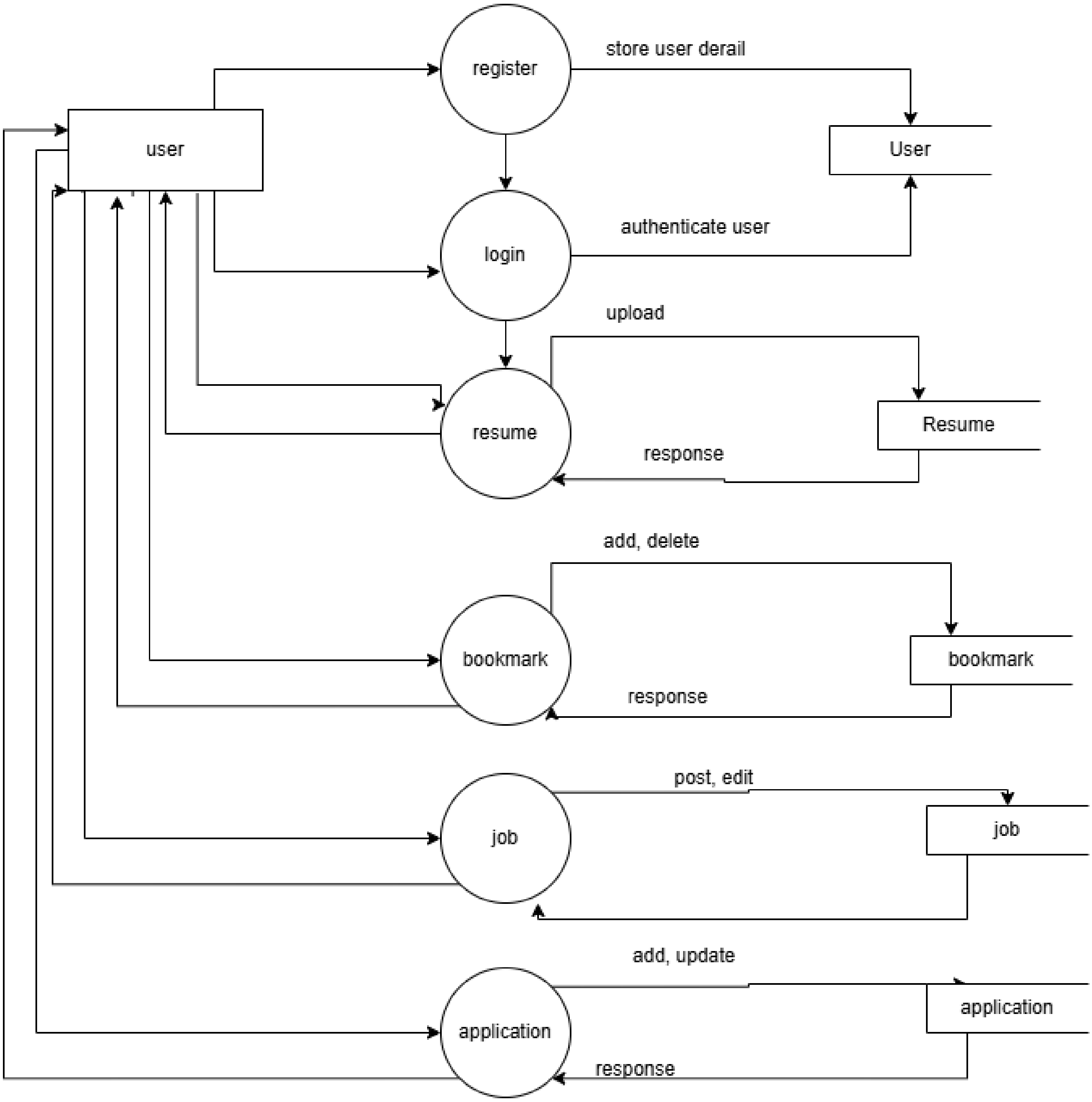


Figure 3.6: Level 1 DFD for Job Portal Web Application

3.2 System Design

To implement the different functional requirements of the system in graphical form, different design diagrams of the system have been prepared which are as follows:

3.2.1 Architectural Design

For this system, three tier architecture is implemented which includes user interface, web server and database. In architectural design, basic structure of the system is shown:

3.2.2 System Flowchart

Flowchart is the pictorial representation of the program logic of the system. The below flowchart represents the system

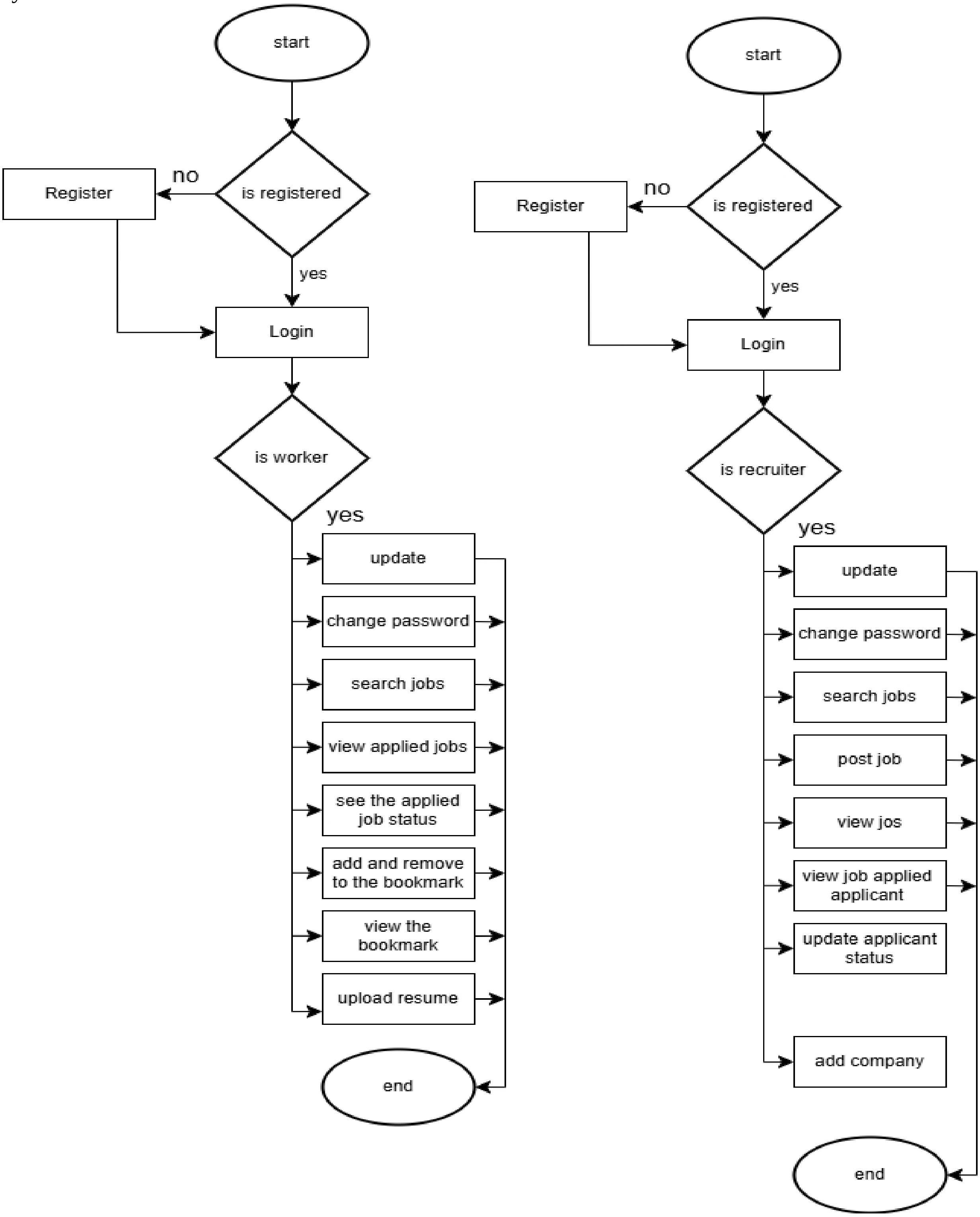


Figure 3.8: System flowchart for Job portal Web Application

3.2.3 Database Schema Design

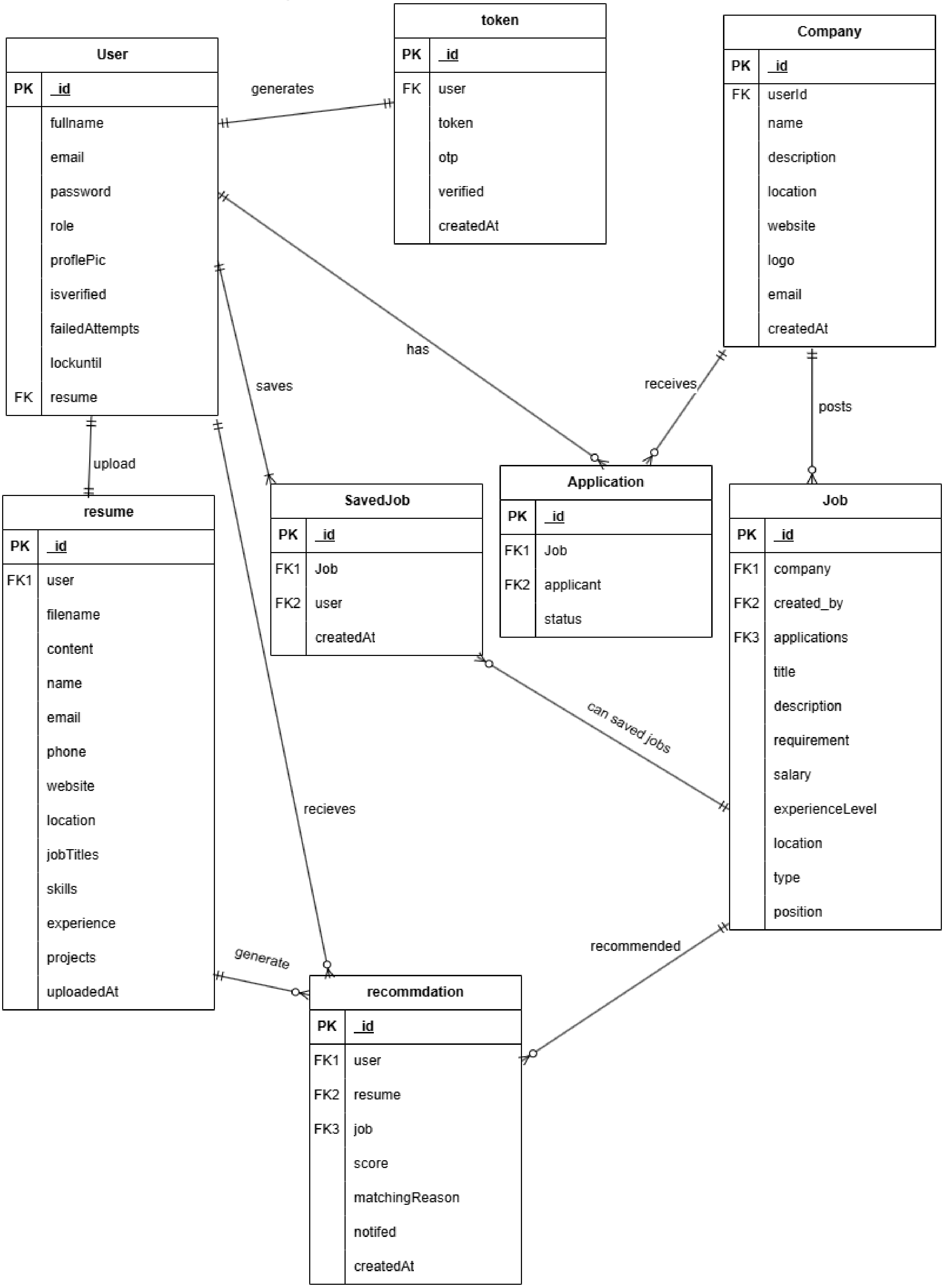


Figure 3.9: Schema design for Job Portal web application

User Interface Design

User interface is one of the most important parts of the system as it determines how easy it is for a new user using the system to understand the different components listed and navigate through them in order to achieve the intended goal of using the system. The interface of our system will be presented according to the following representations:

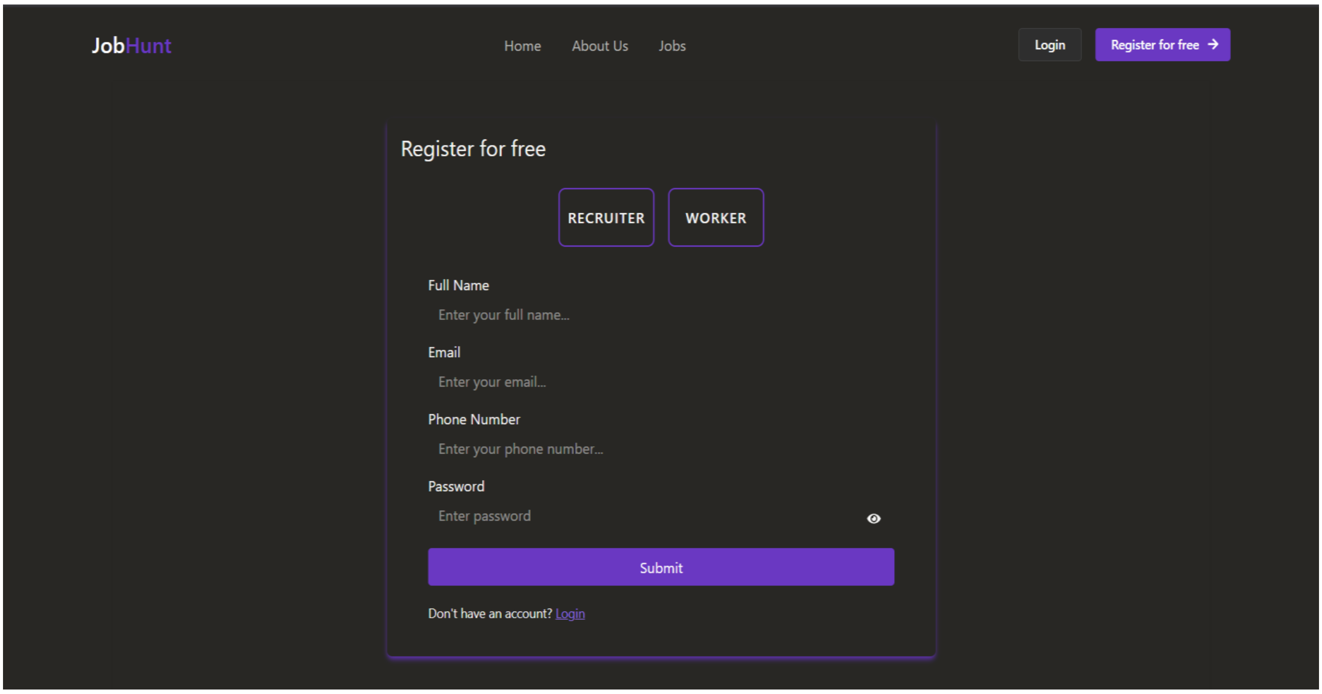


Figure 3.10: Interface for user register

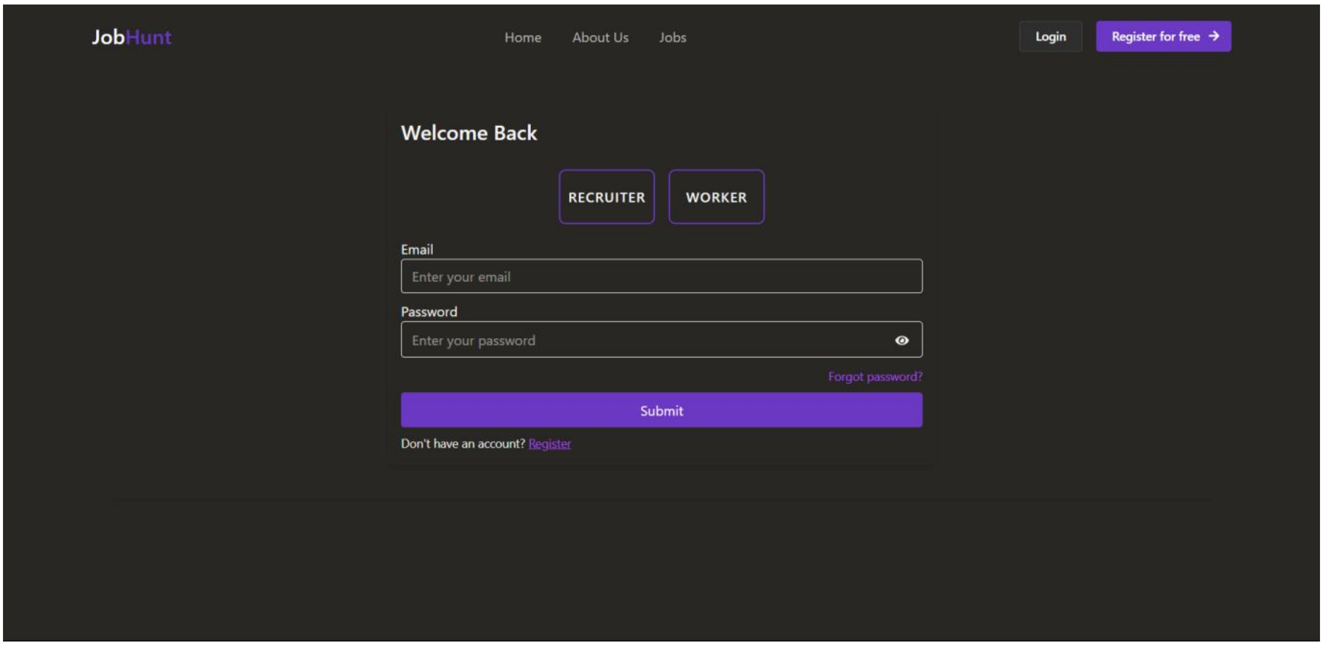
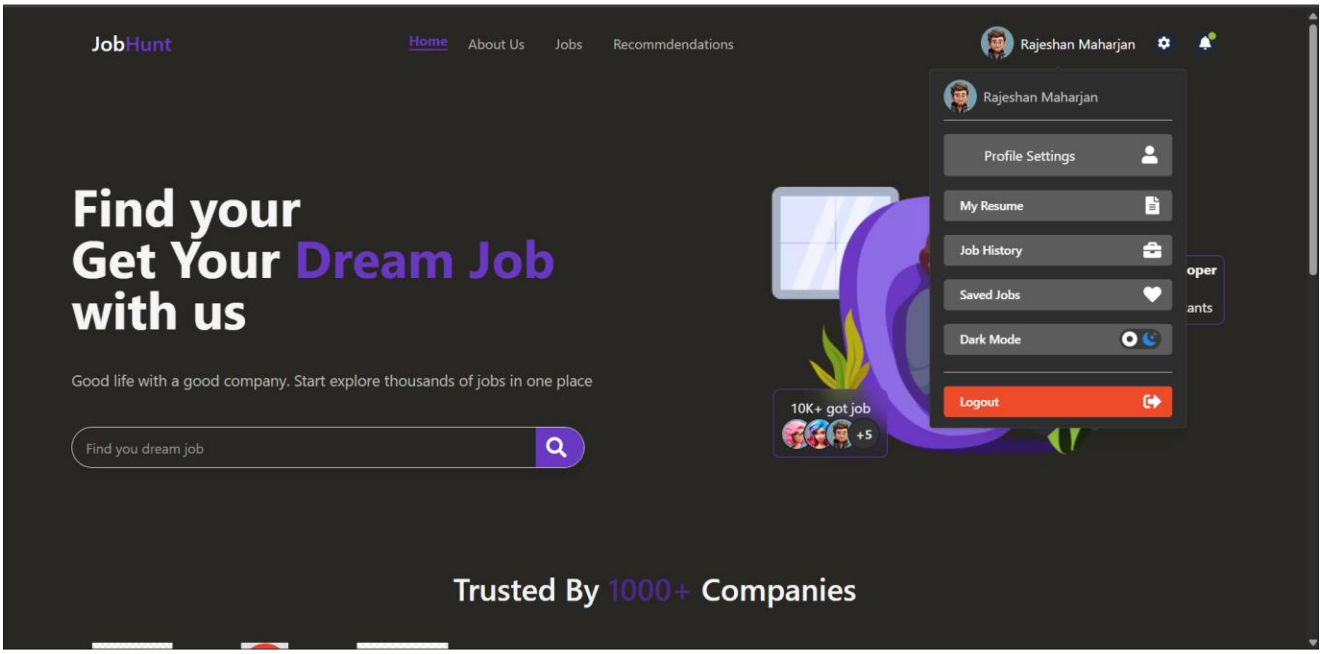


Figure 3.11: Interface for user login

Figure 3.12: Interface for worker



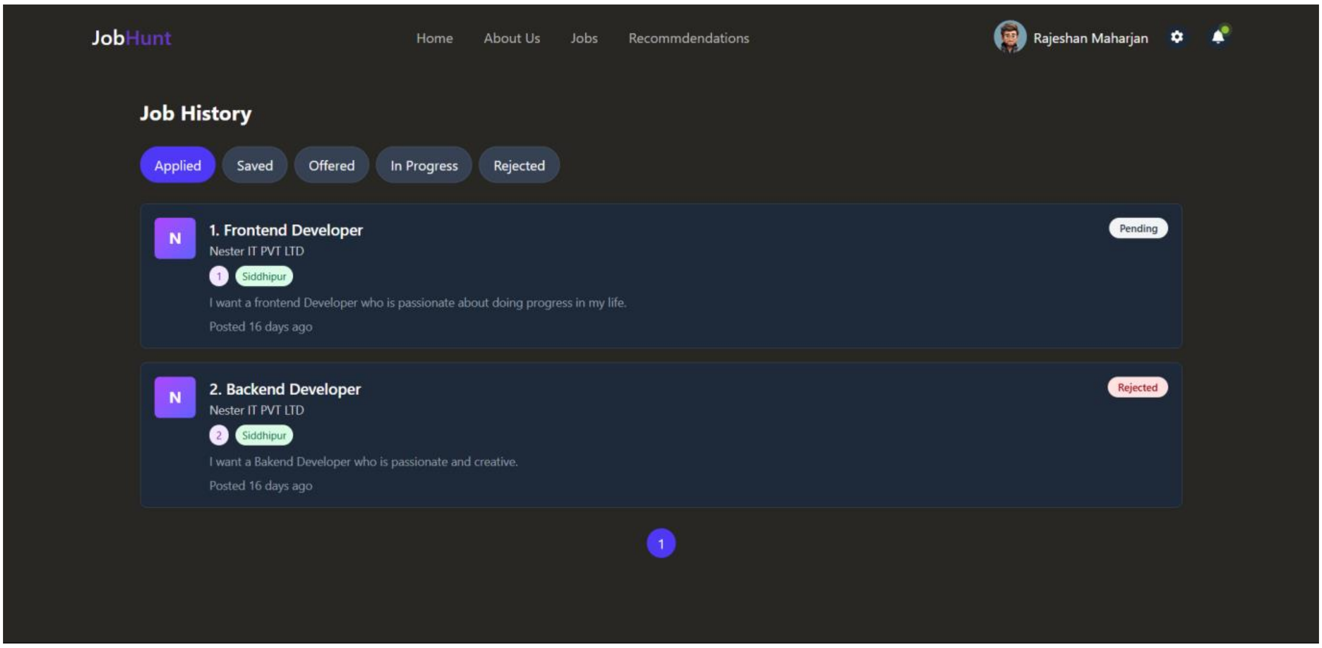


Figure 3.13: Interface for job history

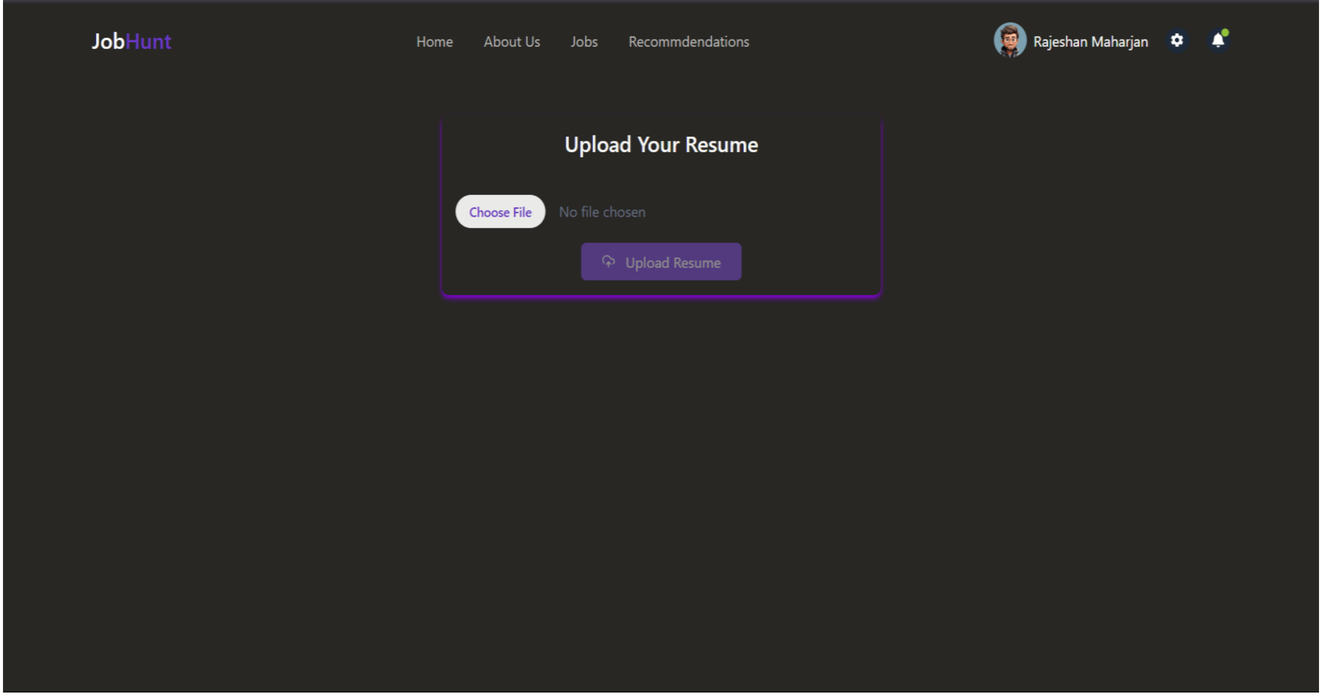


Figure 3.14: Interface for upload resume

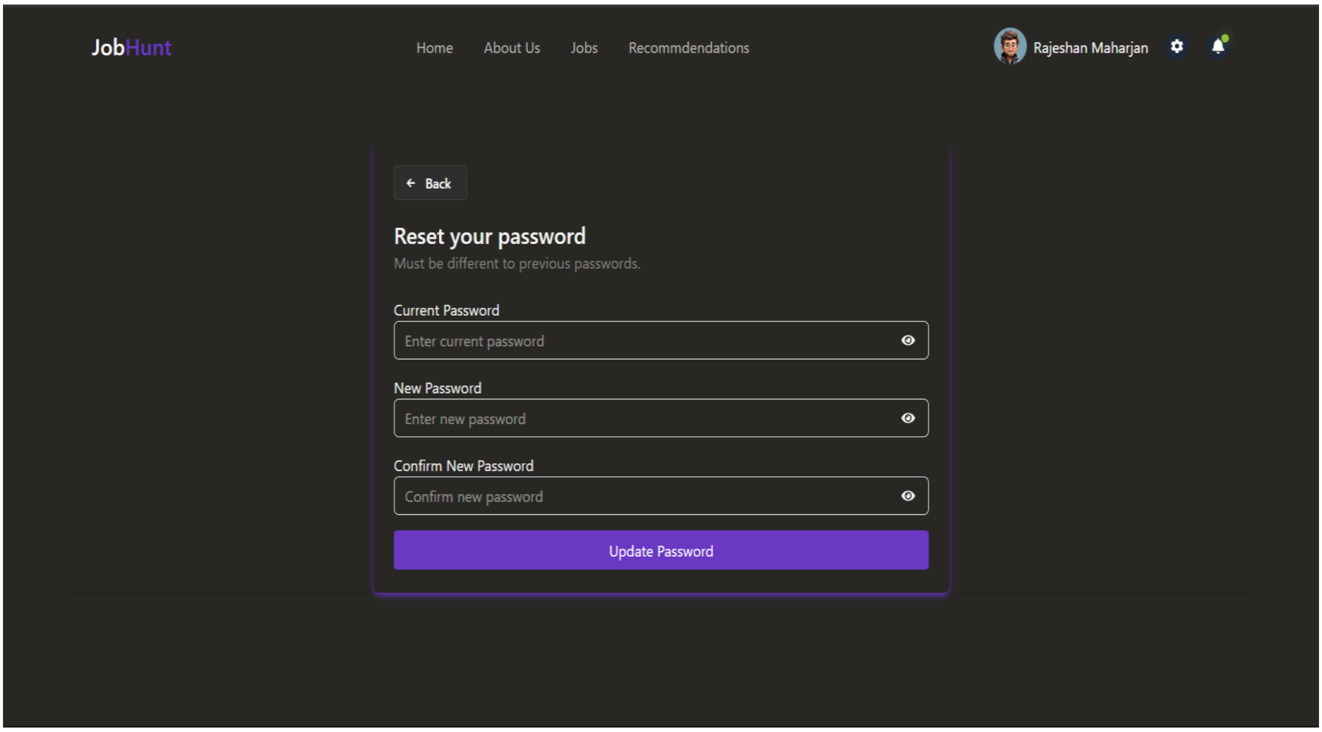


Figure 3.15: Interface for user password reset

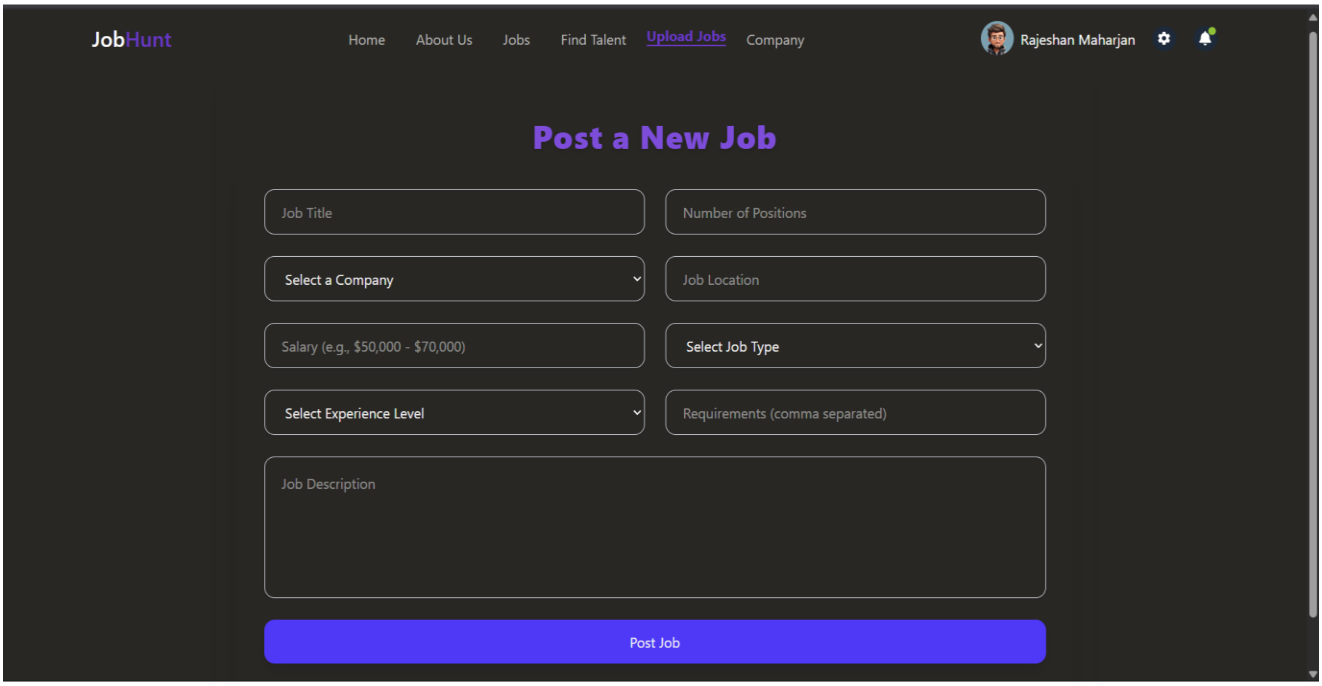


Figure 3.16: Interface for post new job(recruiter)

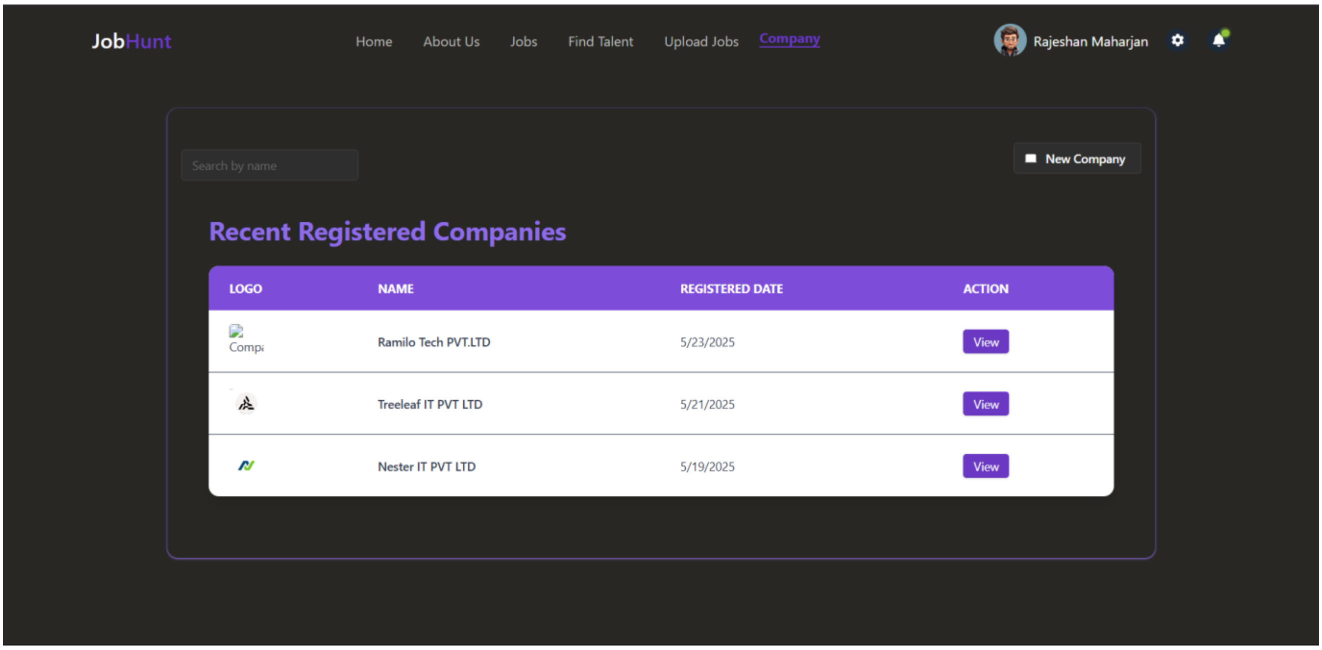


Figure 3.17: Interface for post new job(recruiter)

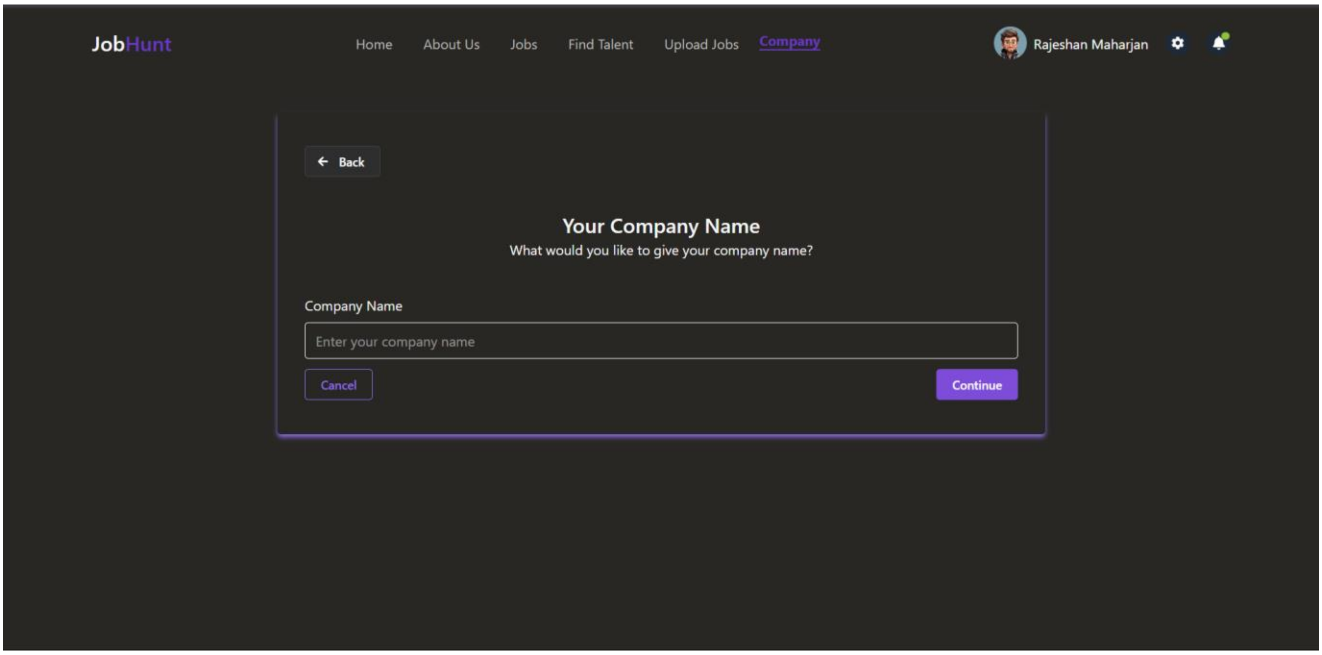


Figure 3.18: Interface for post new job(recruiter)

3.3 Algorithms Details

3.3.1 Resume Parsing and Job Recommendation Algorithm

The Resume Parsing and Job Recommendation Algorithm is designed to automatically extract relevant information from user-uploaded resumes and suggest suitable job listings. The algorithm processes resumes in PDF and DOCX formats, extracts key details such as name, email, skills, experience, and job titles, and matches them against job postings to generate personalized recommendations.

The resume text is extracted using specialized libraries—pdf.js-extract for PDFs and mammoth for DOCX files. After text extraction, the data is cleaned and parsed using pattern matching and keyword recognition techniques. The algorithm calculates similarity scores between the resume content and job descriptions using the Jaccard similarity index, which measures the overlap of keywords and phrases.

Jobs are ranked based on a weighted score incorporating job title matches, skill overlaps, location proximity, and textual similarity. This approach provides efficient and accurate job recommendations tailored to each user's qualifications and preference

CHAPTER 5: CONCLUSION AND RECOMMENDATION

5.1 Lesson Learned/Outcome

Developing the job portal web application significantly enhanced my knowledge and skills in various technical areas. I gained a deep understanding of how to implement a custom keyword matching algorithm for effective job recommendations. Implementing debouncing in the search box taught me how to optimize user input handling to provide fast and accurate data filtering.

Through this project, I learned to develop these algorithms using JavaScript, which helped me strengthen my core programming concepts such as array manipulation, mathematical functions, and data processing techniques. Besides algorithm implementation, I also gained valuable insights into designing a user-friendly interface, managing data storage efficiently, and enhancing user interactivity on a web application.

Furthermore, I learned important backend functionalities, including implementing a secure forgot password flow using OTP and reset links, integrating nodemailer for email verification, and adding features like job bookmarking for improved user experience.

Overall, this project gave me practical experience with full-stack web development, deepened my understanding of key programming concepts, and improved my ability to build robust, user-centric web applications.

5.2 Conclusion

The development of the job portal addresses the challenges faced by job seekers and employers in the traditional job search and recruitment process. By integrating automated resume parsing and intelligent job recommendations, the system enhances the efficiency and accuracy of matching candidates with suitable job opportunities. This platform offers a user-friendly, secure, and effective solution tailored for Nepali users, reducing the time and effort involved in manual job searches. With further improvements and additional features, the job portal has the potential to significantly contribute to the employment landscape.

5.3 Future Recommendations

Job portal web application still lacks some of the necessary features which can implemented in future. Some of the things that can be done to make job portal more better are given below:

- Create a multiple details as per the user
- Can add the ATS score checker.

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REFERENCES

[1] N. Sharma, “What is a Job Portal? Features and Benefits Explained,” *CareerNepal.com*, Accessed: Feb. 20, 2025. [Online]. Available: https://www.careernepal.com/articles/job-portal-definition

[2] M. Rouse, “What is LinkedIn and How It Works,” *Techopedia*, Accessed: Feb. 22, 2025. [Online]. Available: https://www.techopedia.com/definition/28403/linkedin

[3] A. Joshi, “Improving Recruitment Portals with Intelligent Resume Matching,” Journal of Digital Employment Systems, vol. 12, no. 3, pp. 45–52, 2023. Accessed: Feb. 20, 2025. [Online]. Available: https://www.jdigitalemployment.org/articles/improving-portal-matching